Name : Jayashree Muthukumaran
Class : IOT A
Reg No : 2301110 2033

classmate
Date
Page

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★ Gronsfeld cipher with hash:

→ Gronsfeld cipher encryption:

Input : Plaintext + key
Output : Ciphertext

Assumptions
1) only dealing with capital letters.
2) Special other characters will be appended as it is.

```
alpha_count ← count of alphabetic characters
              in plaintext
while len(key) < alpha_count
        key = key + key
R ← " "  (ciphertext)
j ← 0
for i ← 0 to len(plaintext)-1:
    c ← T[i]
    if c is alphabetic then
        shift ← int(k[j])
        j ← j+1
        base = ord('A')
        R ← R + chr((ord(c) - base +
                shift) mod 26)
    else
        R ← R + c
return R
```

→ For decryption:

Input : Ciphertext + key
Output : Plaintext

```
alpha_count ← count of alphabetic characters in
              ciphertext
while (len(key)) < alpha_count:
        key = key + key
~~plaintext~~ R = " "  (plaintext)
```

```
j <- 0
for i <- 0 to len(ciphertext) - 1:
        p <- T[i]
        if p is alphabetic then:
            shift <- int(k[j])
            j <- j + 1
            base = ord('A')
            R <- R + chr((ord(p) - base -
                shift) mod 26 + base)
        else
            R <- R + c
return R
```

(A) Hash function to be used → DJB2 - C (with some modifications to work well with the cipher)

INPUT : text (string)
OUTPUT : hash_digest (alphabetic_string)

```
h <- 5381
magic_multiplier <- 33
for each character c in text:
        h <- h * magic_multiplier + ord(c)
hash_digest <- " ".
for each digit in DIGITS(Str(h)):
        hash_digest <- hash_digest + chr(
                Integer(n) + ord('A'))
return hash_digest.
```

Authentication scheme

- sender(msg, key)

  INPUT: msg(str), key(numeric str)

  OUTPUT: cipher (encrypted authenticated packet)

```
msg ← UPPERCASE(msg)
h ← DJB2-Hash(msg)        // Hashing
packet ← msg + "|" + h
cipher ← gronsfeld_encrypt(packet, key)
                          // Encrypting
return cipher.
```

- receiver(cipher, key)

  INPUT: cipher (encrypted packet), key (numeric string)

  OUTPUT: "Authentic" or "Tampered"

```
packet ← gronsfeld_decrypt(cipher, key)// Decrypting
if "|" NOT in packet:
    return "tampered"
msg, h ← SPLIT(packet, "|")
h_new ← DJB2-Hash(msg)
if h_new == h'   // Checking the authenticity
    return "Authentic"
else
    return "tampered"
```

Overall time complexity for the authentication scheme:

Component	Time Complexity
Encryption	$O(n)$
Decryption	$O(n)$
Hashing	$O(n)$
Full Authentication	$O(n)$